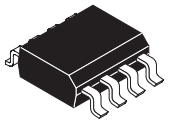
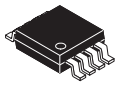
TEXAS INSTRUMENTS





SLAS293 − DECEMBER 2001

5-V, LOW POWER, 16-BIT, 200-KSPS SERIAL ANALOG-TO-DIGITAL CONVERTERS WITH AUTO-POWER DOWN

FEATURES

- 200-KSPS Sampling Rate
- Built-In Conversion Clock
- INL: ±2.5 LSB Max, DNL: 2 to −1 LSB Max
- SINAD = 84.5 dB, SFDR = 95 dB, THD = 94 dB at 15 kHz f_{in}, 200 KSPS
- SPI/DSP-Compatible Serial Interfaces With SCLK Input up to 15 MHz
- Single 5-V Supply
- Rail-to-Rail Analog Input With 500 kHz BW
- Two Input Options Available:
  - TLC4541 − Single Channel Input
  - TLC4545 − Single Channel, Pseudo-differential Input
- (TLC4541) Optimized DSP Interface − Requires FS Input Only
- Low Power With Auto-Power Down
  - Operating Current: 3.5 mA
  - Auto-Power Down Current: 5 μA
- Pin Compatible 12/14/16-Bit Family in 8-Pin SOIC and MSOP Packages

APPLICATIONS

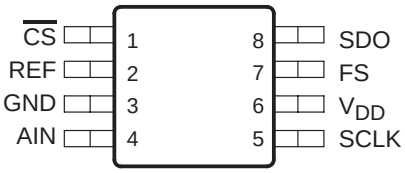
- ATE System
- Industrial Process Control
- Measurement
- Motor Control

DESCRIPTION

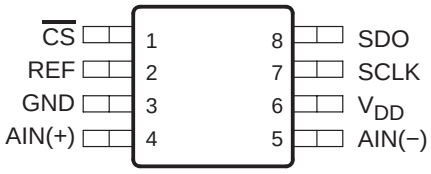
The TLC4541 and TLC4545 are a family of high performance, 16-bit, low power, miniature CMOS analog-to-digital converters (ADCs). These devices operate from a single 5-V supply. Devices are available with single, dual, or single pseudo-differential inputs. All of these devices have a chip select ($\overline{CS}$), serial clock (SCLK), and serial data output (SDO) that provides a direct 3-wire interface to the serial port of most popular host microprocessors (SPI interface). When interfaced with a DSP, a frame sync signal (FS) is used to indicate the start of a serial data frame on either pin 1 ($\overline{CS}$) or pin 7 (FS) for the TLC4541. The TLC4545 ADC connects to the DSP via pin 1 only ($\overline{CS}$).

The TLC4541 and TLC4545 are designed to operate with low power consumption. The power saving feature is further enhanced with an auto-power down mode. This product family features a high-speed serial link to modern host processors with an external SCLK up to 15 MHz. Both families use a built-in oscillator as the conversion clock, providing a 2.94 μs maximum conversion time.

TLC4541
D OR DGK Package
(TOP VIEW)



TLC4545
D OR DGK Package
(TOP VIEW)



Please be aware that an important notice concerning availability, standard warranty, and use in critical applications of Texas Instruments semiconductor products and disclaimers thereto appears at the end of this data sheet.

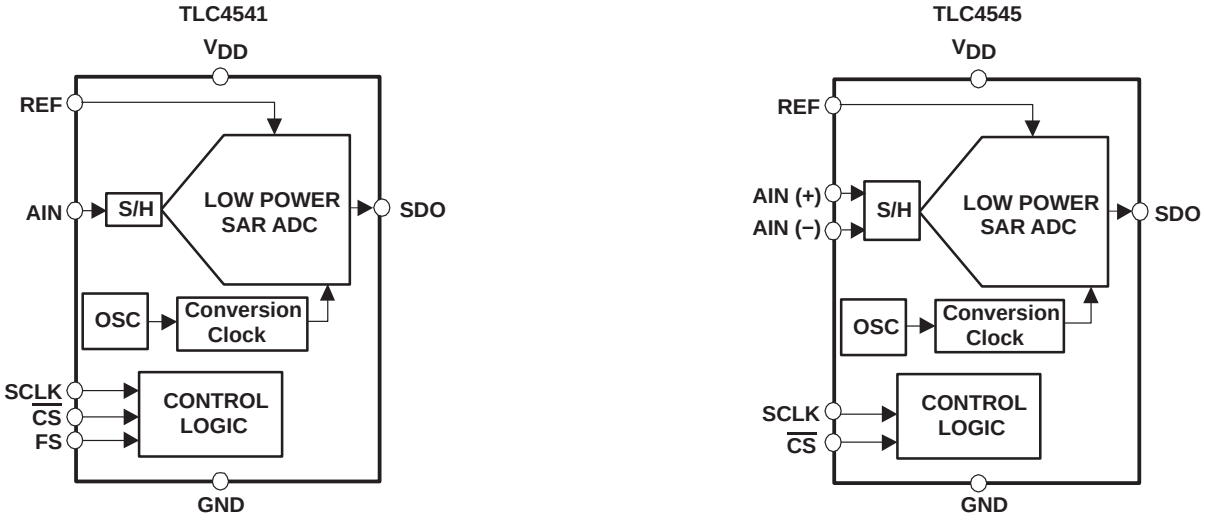
PRODUCTION DATA information is current as of publication date. Products conform to specifications per the terms of Texas Instruments standard warranty. Production processing does not necessarily include testing of all parameters.

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AVAILABLE OPTIONS

T_A	PACKAGED DEVICES	
	8-MSOP (DGK)	8-SOIC (D)
−40°C to 85°C	TLC4541IDGK (PKG Code = ALM)	TLC4541ID
	TLC4545IDGK (PKG Code = AME)	TLC4545ID

functional block diagram



Terminal Functions

TLC4541 single channel unipolar ADCs

TERMINAL NAME	TERMINAL NO.	I/O	DESCRIPTION
AIN	4	I	Analog input channel
$\overline{CS}$	1	I	Chip select. A high-to-low transition on the $\overline{CS}$ input removes SDO from a high-impedance state within a maximum delay time. If the TLC4541 is attached to a dedicated TMS320 DSP serial port using the FS input, $\overline{CS}$ can be grounded.
FS	7	I	DSP frame sync input. Indication of a start of a serial data frame. A low-to-high transition removes SDO from the high-impedance state and the MSB is presented. Tie this pin to V_{DD} if not used.
GND	3	I	Ground return for the internal circuitry. Unless otherwise noted, all voltage measurements are with respect to GND.
SDO	8	O	The 3-state serial data output for the A/D conversion result. SDO is kept in the high-impedance state when $\overline{CS}$ is high. The output format is MSB first. Remaining data bits are presented on the rirsing edge of SCLK. When FS is not active (FS = 1 at the falling edge of $\overline{CS}$): The MSB is presented on the SDO pin on the falling edge of $\overline{CS}$ after a maximum delay time. Data is valid on each falling edge of SCLK until all data is read. When FS is active (FS = 0 at the falling edge of $\overline{CS}$): The MSB is presented to the SDO output on the rising edge of FS. Data is valid on the falling edge SCLK and changes on the rising edge SCLK (this is typically used with an active FS from a DSP). SDO returns to the high-impedance state after the 17th rising edge on SCLK. If a 17th SCLK cycle is not presented, as is the case when using an SPI host, SDO returns to the high-impedance state on the rising edge of $\overline{CS}$.
SCLK	5	I	Serial clock. This terminal receives the serial SCLK from the host processor.
REF	2	I	External voltage reference input
V_{DD}	6	I	Positive supply voltage

TLC4545 single channel pseudo-differential ADCs

TERMINAL NAME	TERMINAL NO.	I/O	DESCRIPTION
AIN0 (+)	4	I	Positive analog input for the TLC4545.
AIN1 (−)	5	I	Inverted analog input for the TLC4545.
$\overline{CS}$	1	I	Chip select. A high-to-low transition on $\overline{CS}$ removes SDO from the high-impedance state within a maximum delay time. The $\overline{CS}$ input can be connected to a DSP frame sync (FS) output when a dedicated TMS320 DSP serial port is used.
GND	3	I	Ground return for the internal circuitry. Unless otherwise noted, all voltage measurements are with respect to GND.
SDO	8	O	The 3-state serial data output for the A/D conversion result. SDO is kept in the high-impedance state when $\overline{CS}$ is high and presents output data after the $\overline{CS}$ falling edge until the LSB is presented. The output format is MSB first. The remaining data bits are presented on the rising edge of SCLK. Output data is valid on each falling edge of SCLK until all data is read. SDO returns to the high-impedance state after the 17th rising edge on SCLK. If a 17th SCLK cycle is not presented, as is the case when using an SPI host, SDO returns to the high-impedance state on the rising edge of $\overline{CS}$.
SCLK	7	I	Serial clock. This terminal receives the serial SCLK from the host processor.
REF	2	I	External voltage reference input
V_{DD}	6	I	Positive supply voltage

absolute maximum ratings over operating free-air temperature (unless otherwise noted)†

Supply voltage, GND to V_{DD} . . . . . −0.3 V to 6.5V
Analog input voltage range . . . . . −0.3 V to V_{DD}+0.3 V
Reference input voltage . . . . . V_{DD}+0.3 V
Digital input voltage range . . . . . −0.3 V to V_{DD}+0.3 V
Operating virtual junction temperature range, T_J . . . . . −40°C to 150°C
Operating free-air temperature range: T_A (I) . . . . . −40°C to 85°C
Storage temperature range, T_{stg} . . . . . −65°C to 150°C
Lead temperature 1,6 mm (1/16 inch) from case for 10 seconds . . . . . 260°C

† Stresses beyond those listed under “absolute maximum ratings” may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under “recommended operating conditions” is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

recommended operating conditions

		MIN	NOM	MAX	UNIT
Supply voltage, V_{DD}		4.5	5	5.5	V
Frequency, SCLK	V_{DD} = 2.7 V to 5.5 V	100		15000	kHz
Tolerable clock jitter, SCLK	V_{DD} = 2.7 V to 5.5 V			24	ps
Aperature jitter	V_{DD} = 2.7 V to 5.5 V		100		ps
External reference voltage input, V_{REF}		4		V_{DD}	V
V_{REF} input impedance	V_{DD} = 5 V, $\overline{CS}$ = 1, SCLK = 0	100			MΩ
	V_{DD} = 5 V, $\overline{CS}$ = 0, SCLK = 15 MHz	20	25		kΩ
External reference input current	V_{DD} = V_{REF} = 4.5 V, $\overline{CS}$=0, SCLK = 15 MHz		0.02	1	mA
Analog input voltage	AIN, AIN(+)	0		V_{DD}	V
	AIN(−)	−0.2		0.2	
High level control input voltage, V_{IH}		2.1			V
Low level control input voltage, V_{IL}				0.8	V
Operating free-air temperature, T_A	TLC4541/45I	−40		85	°C

electrical characteristics over recommended operating free-air temperature range, V_{DD} = 5 V, V_{REF} = 4.096 V, SCLK frequency = 15 MHz (unless otherwise noted)

PARAMETER		TEST CONDITIONS		MIN	TYP	MAX	UNIT
V_{OH}	High-level output voltage	V_{DD} = 4.5 V,	I_{OH} = −0.2 mA	3.9			V
V_{OL}	Low-level output voltage	V_{DD} = 4.5 V,	I_{OL} = 0.8 mA			0.4	V
I_{OZ}	Off-state output current (high-impedance-state)	$V_O = V_{DD}$,	$\overline{CS} = V_{DD}$		1	2.5	µA
		V_O = 0,	$\overline{CS} = V_{DD}$		−1	−2.5	
I_{IH}	High-level input current	$V_I = V_{DD}$			0.005	2.5	µA
I_{IL}	Low-level input current	V_I = 0			−0.005	2.5	µA
I_{CC}	Operating supply current	$\overline{CS}$ at 0 V,	V_{DD} = 4.5 V to 5.5 V			3.5	mA
$I_{CC(PD)}$	Power-down supply current	For all digital inputs, 0≤ V_I ≤ 0.3 V or V_I ≥ V_{DD} − 0.3 V, SCLK=V_{DD}, V_{DD} = 4.5 V to 5.5 V			3	5	µA
	Selected analog input channel leakage current	Selected channel at V_{DD}				1	µA
		Selected channel at 0 V				−1	
C_i	Input capacitance	Analog inputs			11	14	pF
		Control Inputs			20	25	
Z_i	Input resistance	V_{DD} = 5.5 V				500	Ω

ac specifications (TLC4541/45)

PARAMETER			TEST CONDITIONS	MIN	TYP	MAX	UNIT
SINAD	Signal-to-noise ratio + distortion		f_I = 15 kHz at 200 KSPS		84.5		dB
SNR	Signal-to-noise ratio		f_I = 15 kHz at 200 KSPS		85		
THD	Total harmonic distortion	TLC4541	f_I = 15 kHz at 200 KSPS		−94	−87	dB
		TLC4545	f_I = 15 kHz at 200 KSPS		−94	−89	
ENOB	Effective number of bits		f_I = 15 kHz at 200 KSPS		13.7		Bits
SFDR	Spurious free dynamic range	TLC4541	f_I = 15 kHz at 200 KSPS		−95	−87	dB
		TLC4545	f_I = 15 kHz at 200 KSPS		−95	−89	
	Full power bandwidth, −3 dB, analog input				1		MHz
	Full power bandwidth, −1 dB, analog input				500		kHz
Crosstalk			0.25 LBS	80			dB

dc specifications (TLC4541/45)

PARAMETER		TEST CONDITIONS	MIN	TYP†	MAX	UNIT
INL	Integral linearity error (see Note 1)		−2.5		2.5	LSB
DNL	Differential linearity error		−1		2	LSB
E_O	Offset error (see Note 2)	TLC4541	−3.5		3.5	mV
		TLC4545	−1		1	
E_G	Gain error (see Note 2)	TLC4541	−2		2	mV
		TLC4545	−1.8		1.8	

† All typical values are at V_{DD} = 5 V, T_A = 25°C.

NOTES: 1. Linear error is the maximum deviation from the best straight line through the A/D transfer characteristics.
2. Zero error is the difference between 0000h and the converted output for zero input voltage: full-scale error is the difference between ideal full-scale and the converted output for full-scale input voltage.

timing requirements, V_{DD} = 5 V, V_{REF} = 4.096 V, SCLK frequency = 15 MHz (unless otherwise specified)

		MIN	TYP	MAX	UNIT
$t_{cyc(SCLK)}$	SCLK cycle time, V_{DD} = 4.5 V to 5.5 V (see Note 3)	66		10000	ns
t_{w1}	Pulse width, SCLK low	27		5000	ns
t_{w2}	Pulse width, SCLK high	27		5000	ns
t_{h1}	Hold time, $\overline{CS}$ high after SCLK falling edge	3			ns
t_{su1}	Setup time, $\overline{CS}$ falling edge before the first SCLK falling edge	15			ns
t_{h2}	Hold time, $\overline{CS}$ low after 16th SCLK falling edge	5			ns
t_{w3}	Pulse width, $\overline{CS}$ high	0.5			SCLKs
t_{d1}	Delay time, $\overline{CS}$ falling edge to SDO MSB valid, V_{DD} = V_{REF} = 4.5 V, 20 pF		12	17	ns
t_{d2}	Delay time, SCLK rising edge to next SDO data bit valid, V_{DD} = V_{REF} = 4.5 V, 20 pF			15	ns
t_{d3}	Delay time, 17th SCLK rising edge to SDO 3-stated, V_{DD} = V_{REF} = 4.5 V, 20 pF (see Note 4)			20	ns
t_{su3}	Setup time, $\overline{CS}$ falling edge before FS rising edge (TLC4541 only)	0.5	1		SCLKs
t_{w4}	Pulse width, FS high (TLC4541 only)	0.5	1		SCLKs
t_{su4}	Setup time, FS rising edge before SCLK falling edge (TLC4541 only)	12.5			ns
t_{h4}	Hold time, FS high after SCLK falling edge (TLC4541 only)	5			ns
t_{su5}	Setup time, FS falling edge before 1st SCLK falling edge (TLC4541 only)	12			ns
t_{d4}	Delay time, FS rising edge to SDO MSB valid, (V_{DD} = V_{REF} = 4.5 V, 20 pF TLC4541 only)			15	ns
t_{h6}	Hold time, $\overline{CS}$ low after 1st SCLK falling edge	5			ns
t_{su6}	Setup time, $\overline{CS}$ rising edge before 9th (or the last) SCLK falling edge	5			ns
t_{h7}	Hold time, FS low after 1st SCLK falling edge (TLC4541 only)	5			ns
t_{su7}	Setup time, FS rising edge before 9th (or the last) SCLK falling edge	5			ns
$t_{cyc(reset)}$	Active $\overline{CS}$/FS cycle time, SCLK falling edges required to initialize ADC	1		8	SCLKs
t_{conv}	Conversion time (22 conversion clocks based on 7.5-MHz to 12-MHz OSC)	1.83		2.94	µs
t_s	Sample time, 20 SCLKs, SCLK up to 15 MHz	1.33		200	µs

NOTES: 3. Timing specifications given for 40/60 to 60/40 duty cycle

4. SDO goes into the high impedance state after detection of the 17th rising SCLK edge or a rising $\overline{CS}$ edge if a 17th SCLK is not presented.

TYPICAL CHARACTERISTICS

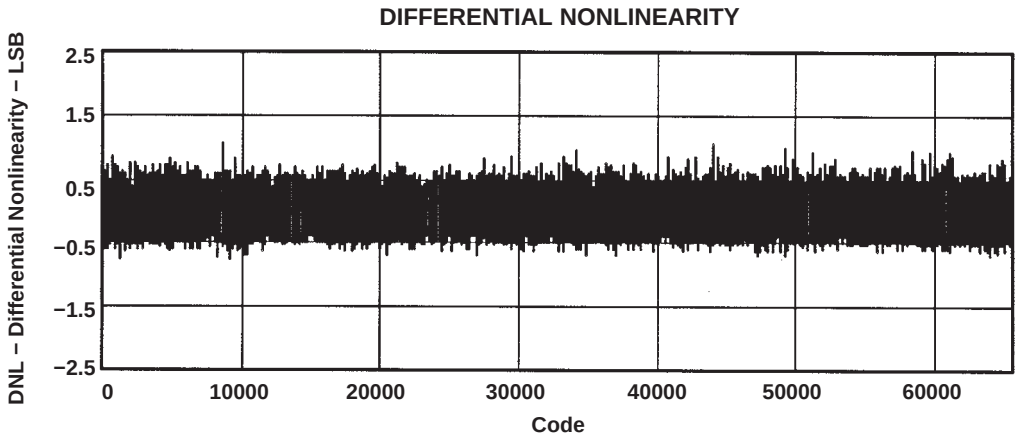


Figure 1

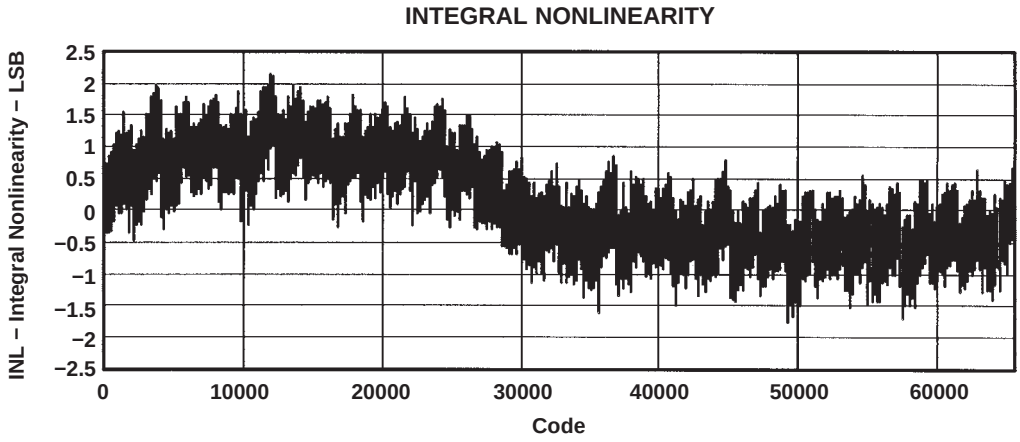


Figure 2

TYPICAL CHARACTERISTICS

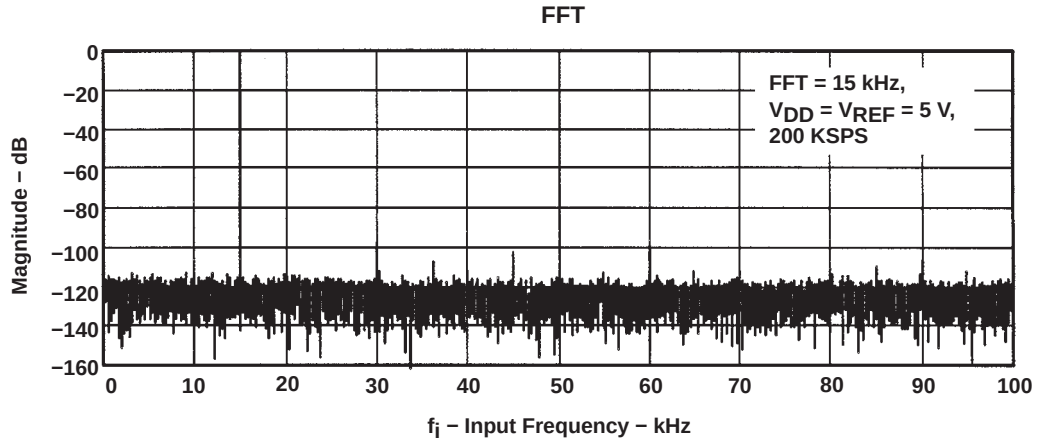


Figure 3

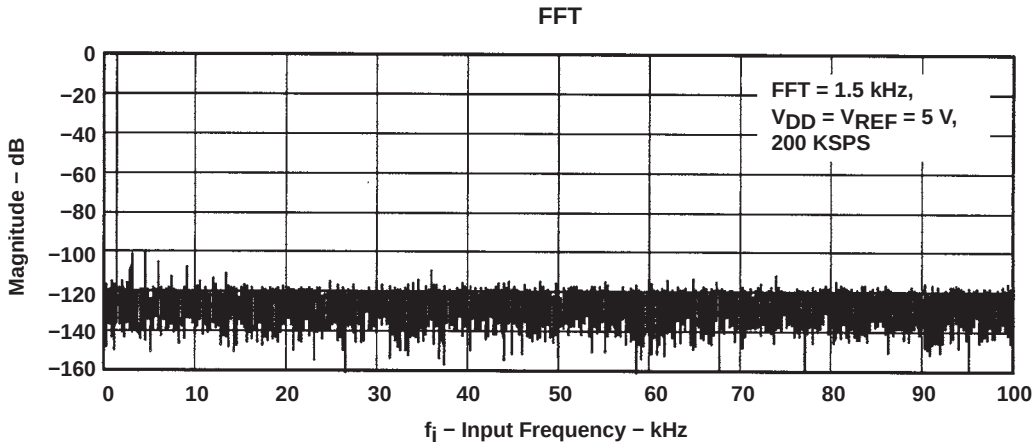


Figure 4

TYPICAL CHARACTERISTICS

SIGNAL-TO NOISE RATIO
vs
INPUT FREQUENCY

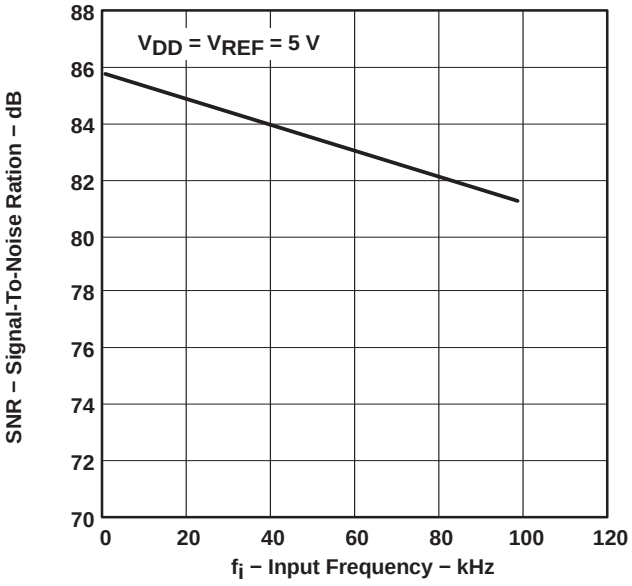


Figure 5

TOTAL HARMONIC DISTORTION
vs
INPUT FREQUENCY

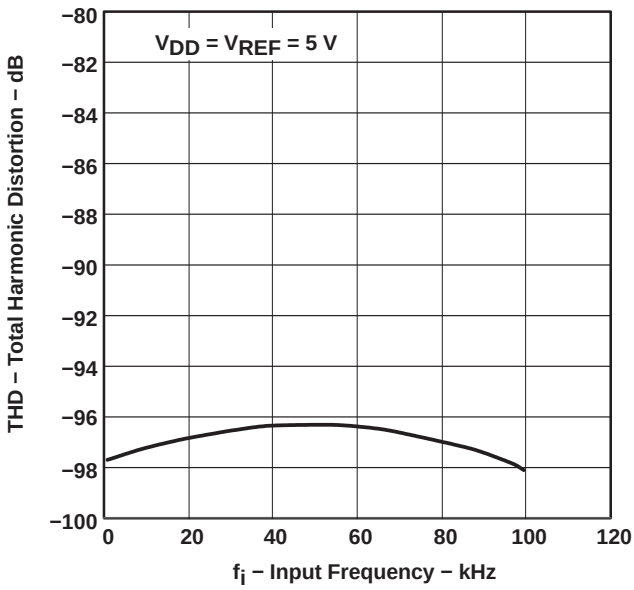


Figure 6

TOTAL HARMONIC DISTORTION
vs
FREE-AIR TEMPERATURE

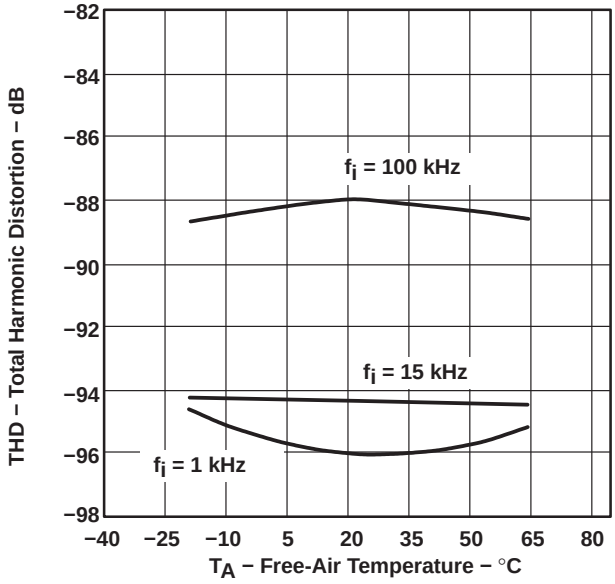


Figure 7

SIGNAL-TO-NOISE RATIO
vs
FREE-AIR TEMPERATURE

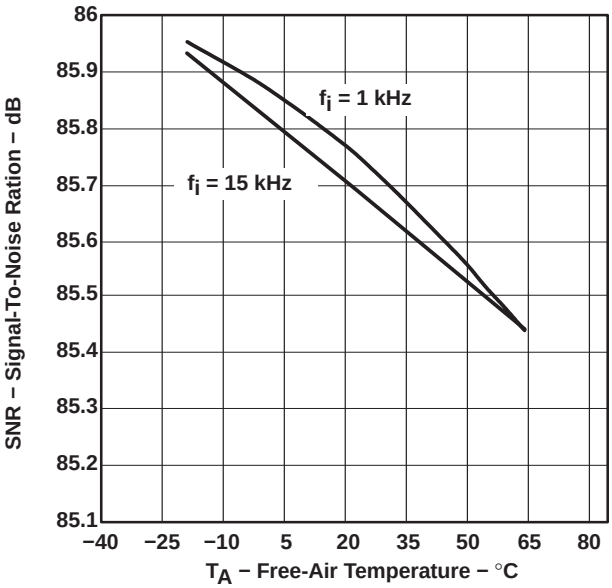


Figure 8

TYPICAL CHARACTERISTICS

TOTAL HARMONIC DISTORTION
vs
REFERENCE VOLTAGE

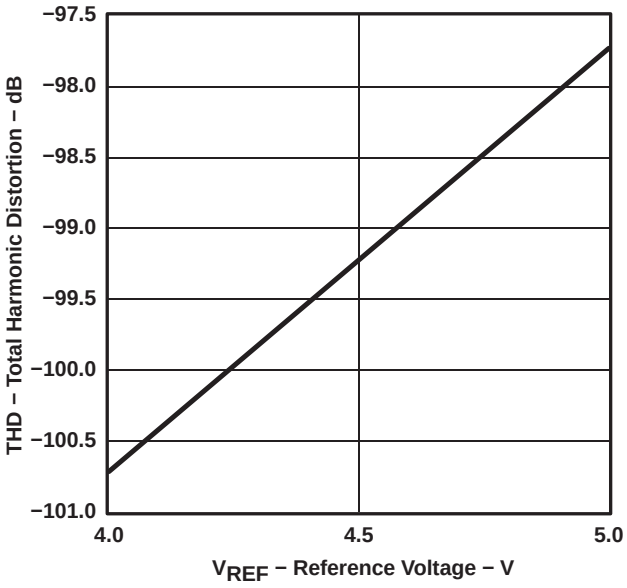


Figure 9

SIGNAL-TO-NOISE RATIO
vs
REFERENCE VOLTAGE

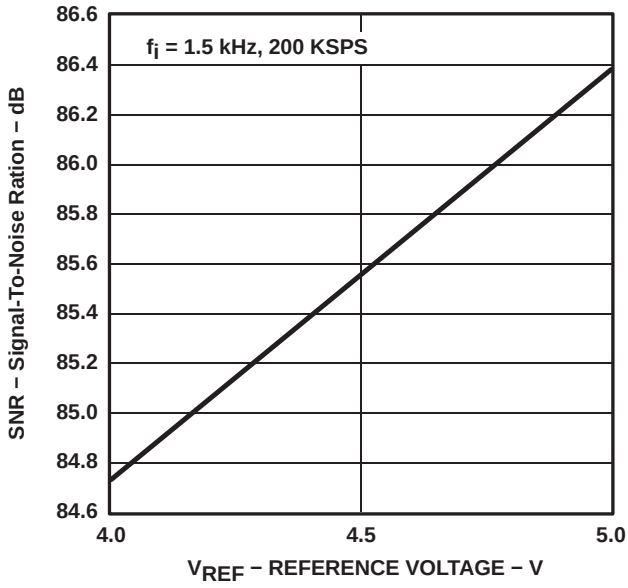


Figure 10

MAXIMUM DIFFERENTIAL NONLINEARITY
vs
REFERENCE VOLTAGE

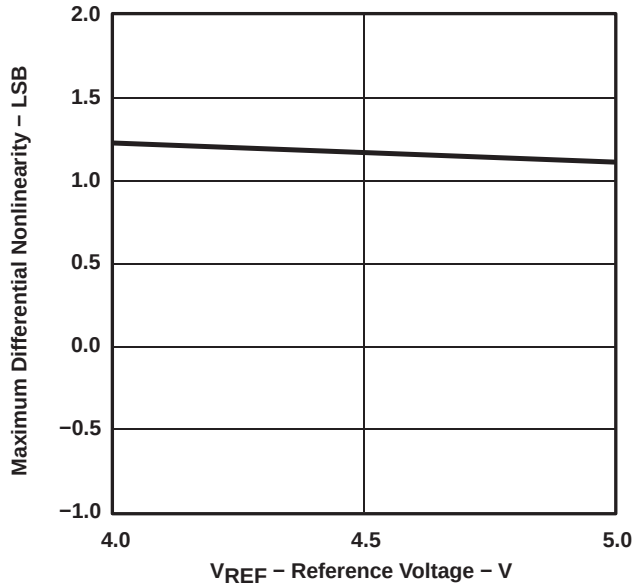


Figure 11

MINIMUM DIFFERENTIAL NONLINEARITY
vs
REFERENCE VOLTAGE

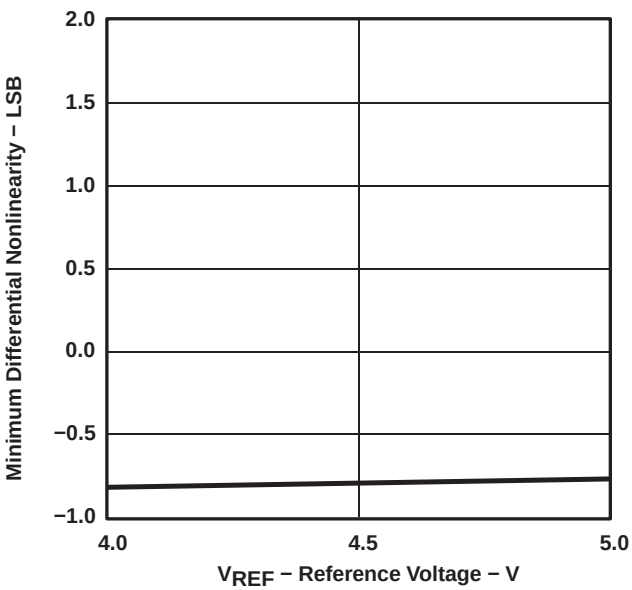


Figure 12

TYPICAL CHARACTERISTICS

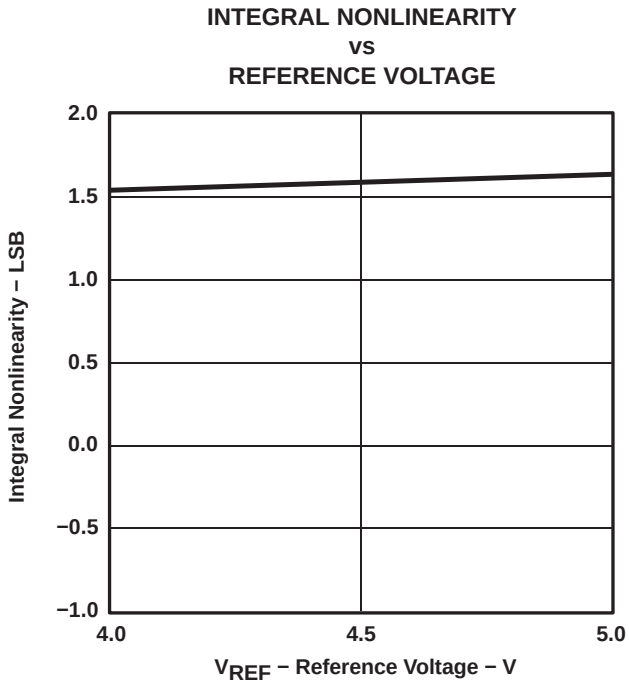


Figure 13

PRINCIPLES OF OPERATION

control and timing

device initialization/RESET cycle

The TLC4541/45 each require one RESET cycle after power-on for initialization in order to operate properly. This RESET cycle is initiated by asserting the $\overline{CS}$ pin (pin 1) low for a minimum duration of at least one SCLK falling edge but no more than 8 SCLK falling edges in length. The RESET cycle is terminated by asserting $\overline{CS}$ high. If a valid RESET cycle is issued, the data presented on the SDO output during the following cycle is FF00h. This output code is useful in determining when a valid reset/initialization has occurred.

The TLC4541 has separate $\overline{CS}$ and FS pins. In this case, it is also possible to initiate the RESET cycle by asserting FS low if $\overline{CS}$ is already low. The RESET cycle can be terminated by either asserting $\overline{CS}$ high (as shown in the first RESET cycle in Figure 14), or by asserting FS high (as shown in the second RESET cycle in Figure 14), whichever happens first.

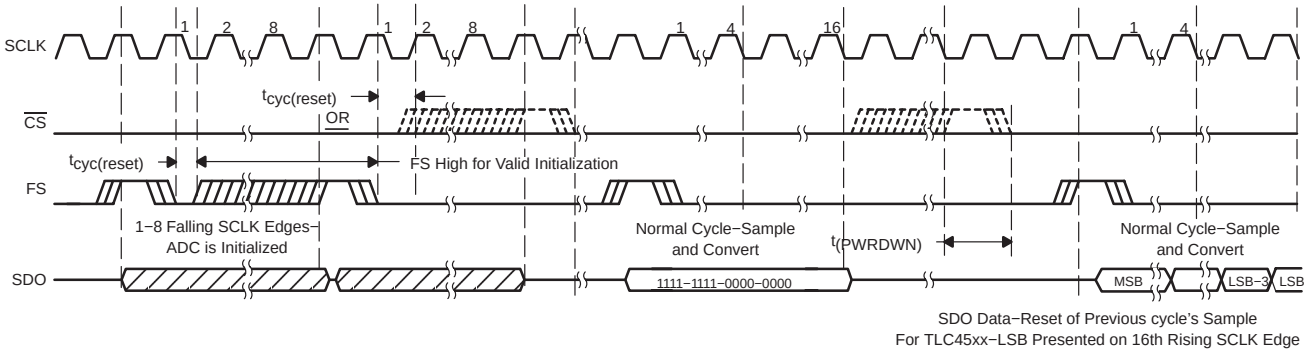


Figure 14. TLC4541/45 Initialization Timing

sampling

The converter sample time is 20 SCLKs in duration, beginning on the 5th SCLK received during an active signal on the $\overline{CS}$ input (or FS input for the TLC4541.)

conversion

Each device completes a conversion in the following manner. The conversion is started after the 24th falling SCLK edge. The $\overline{CS}$ input can be released at this point or at any time during the remainder of the conversion cycle. The conversion takes a maximum of 2.94 µs to complete. Enough time (for conversion) should be allowed before the next falling edge on the $\overline{CS}$ input (or rising edge on the FS input for the TLC4541) so that no conversion is terminated prematurely. If the conversion cycle is terminated early, the data presented on SDO during the following cycle is FF00h. This predefined output code is helpful in determining if the cycle time is not long enough to complete the conversion. The same code is also used to determine if a reset cycle is valid.

For all devices, the SDO data presented during a cycle is the result of the conversion of the sample taken during the previous cycle. The output data format is shown in the following table.

SERIAL OUTPUT DATA FORMAT		
	MSB [D15:D2]	**LSB [D1:D0]**
TLC4541/45	Conversion Result (OD15−OD2)	Conversion Result (OD1 − OD0)

PRINCIPLES OF OPERATION

control and timing (continued)

sampling and conversion cycle

TLC4541:

Control via pin 1, $\overline{CS}$ (FS = 1 at the falling edge of $\overline{CS}$)− The falling edge of $\overline{CS}$ is the start of the cycle. Transitions on $\overline{CS}$ can occur when SCLK is high or low. The MSB may be read on the first falling SCLK edge after $\overline{CS}$ is low. Output data changes on the rising edge of SCLK. This control method is typically used for a microcontroller with an SPI interface, although it can also be used for a DSP. The microcontroller SPI interface should be programmed for CPOL=0 (serial clock inactive low) and CPHA=1 (data valid on the falling edge of serial clock).

Control via pin 7, FS ($\overline{CS}$ is tied/held low)− The rising edge of FS is the start of the cycle. Transitions on FS can occur when SCLK is high or low. The MSB is presented on SDO after the rising edge of FS. The MSB may be read on the first falling edge of SCLK after the FS falling edge. Output data changes on the rising edge of SCLK. This is the typical configuration when the ADC is the only device on the TMS320 DSP serial port.

Control via pin 1 and pin 7, $\overline{CS}$ and FS− Transitions on $\overline{CS}$ and FS can occur when SCLK is high or low. The MSB is presented after the rising edge of FS. The falling edge of FS is the start of the sampling cycle. The MSB may be read on the first falling edge of SCLK after the FS falling edge. Output data changes on the rising edge of SCLK. This is typically used for multiple devices connected to a single TMS320 DSP serial port.

TLC4545:

All control is provided using the $\overline{CS}$ input (pin 1) on the TLC4545. Transitions on $\overline{CS}$ can occur when SCLK is high or low. The cycle is started on the falling edge transition on the $\overline{CS}$ input. This signal can be provided by either a $\overline{CS}$ signal (when interfacing with an SPI microcontroller) or FS signal (when interfacing with a TMS320 DSP). The MSB is presented to SDO on the falling edge of the signal applied to pin 1 and may be read on the first falling SCLK edge after this input is low. Output data changes on the rising edge of SCLK.

control modes

control via pin 1 ($\overline{CS}$, SPI interface)

All devices are compatible with this mode of operation. A falling edge on the $\overline{CS}$ input initiates the cycle. (For the TLC4541, the FS input is tied to V_{DD}). The $\overline{CS}$ input remains low for the entire sampling time plus 4 SCLK decoding time(16 falling SCLK edges) and can then be released at any point during the remainder of the conversion. Enough time should be allowed before the next falling $\overline{CS}$ edge so that the conversion cycle is not terminated prematurely. The microcontroller SPI interface should be programmed for CPOL=0 (serial clock inactive low) and CPHA=1 (data is valid on the falling edge of serial clock).

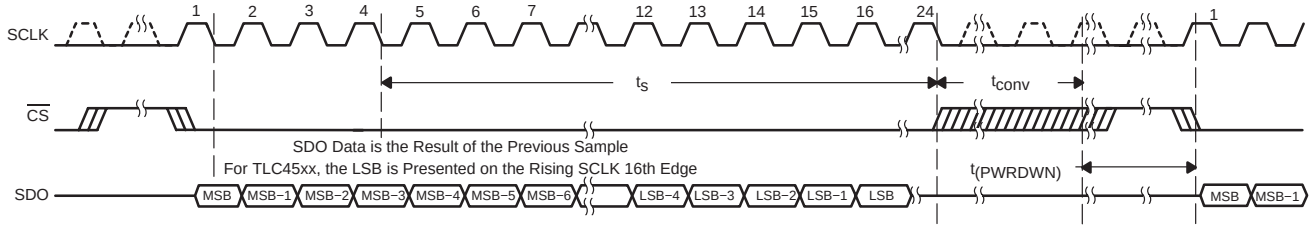


Figure 15. SPI Cycle TIming Using the CS Input (FS = 1 for TLC4541)

PRINCIPLES OF OPERATION

control via pin 1 ($\overline{CS}$, DSP interface)

All devices are compatible with this mode of operation. The FS signal from a DSP is connected directly to the $\overline{CS}$ input of the ADC. A falling edge on the $\overline{CS}$ input while SCLK is high or low initiates the cycle. (For TLC4541 in this configuration, the FS input is tied to V_{DD}.) Enough time should be allowed before the next rising $\overline{CS}$ edge so that the conversion cycle is not terminated prematurely.

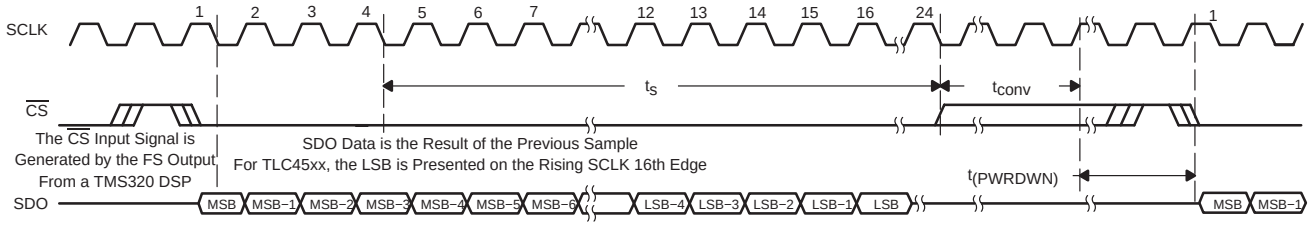


Figure 16. DSP Cycle Timing Using the CS Input (FS = 1 for TLC4541 Only)

control via pin 1 and pin 7 ($\overline{CS}$ and FS or FS only, DSP interface)

Only TLC4541 is compatible with this mode of operation. The $\overline{CS}$ input to the ADC can be controlled via a general-purpose I/O pin from the DSP or tied to ground. The FS signal from the DSP is connected directly to the FS input of the ADC. A rising FS edge releases the MSB to the SDO output. The falling edge on the FS input while SCLK is high or low initiates the cycle. The $\overline{CS}$ input should remain low for the entire sampling time plus 4 SCLK decoding time after falling FS (24 falling SCLK edges) and can then be released at any time during the remainder of the conversion cycle. The optimum DSP interface is achieved when tying $\overline{CS}$ to ground and using only the FS input to control the ADC.

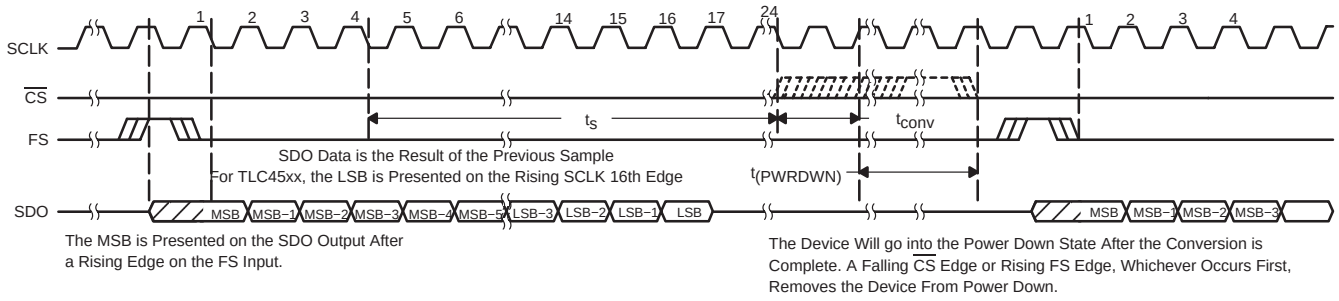


Figure 17. DSP Cycle Timing Using FS Only (or Using Both $\overline{CS}$ and FS for the TLC4541)

PRINCIPLES OF OPERATION

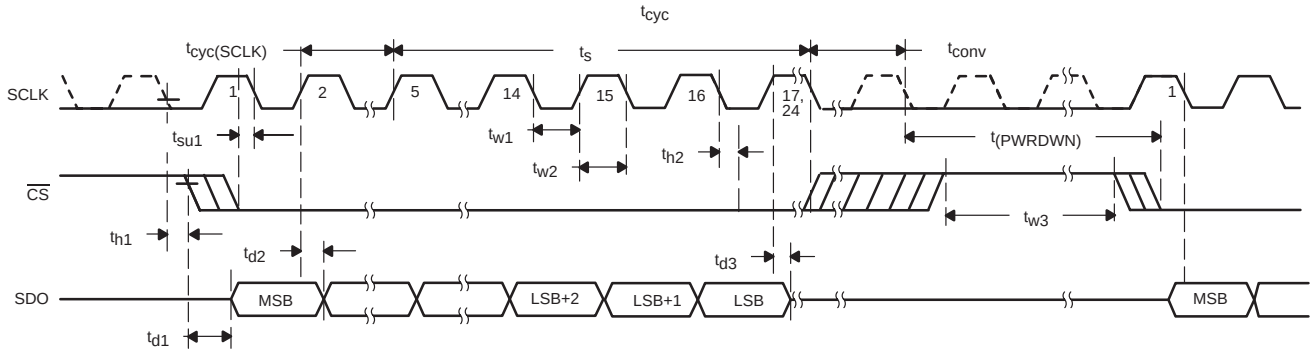


Figure 18. Critical Timing: Control Via CS Input (FS = 1 for TLC4541)

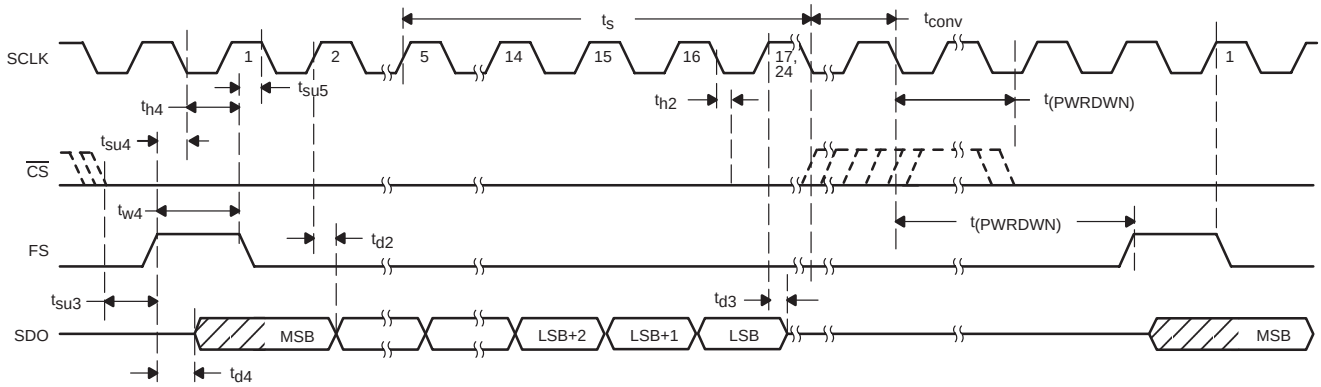


Figure 19. Critical Timing: Control Via CS and FS Inputs (TLC4541 Only)

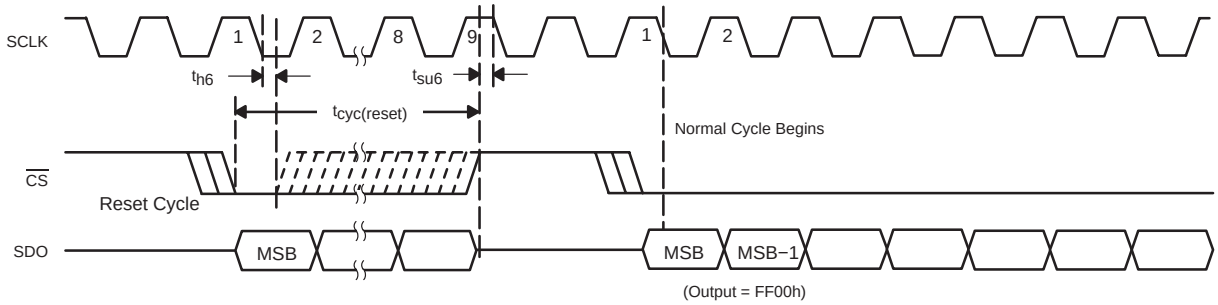


Figure 20. Critical Timing: Reset/Initialization Cycle (FS =1 for TLC4541)

PRINCIPLES OF OPERATION

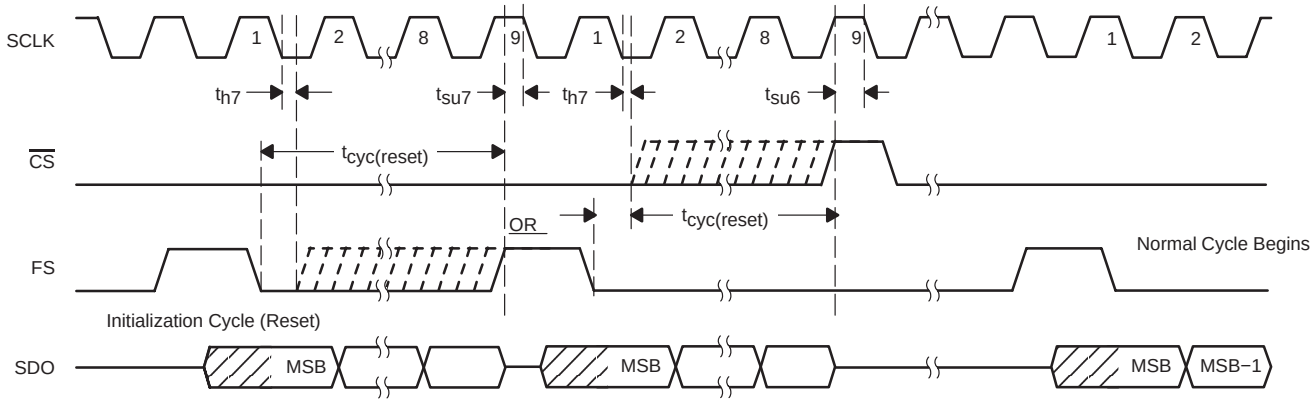


Figure 21. Critical Timing: Initialization Cycle (TLC4541 Only)

detailed description

The TLC4541/5 are successive approximation (SAR) ADCs utilizing a charge-redistribution DAC. Figure 22 shows a simplified version of the ADC. The sampling capacitor acquires the signal on AIN (or the AIN(+) pin for TLC4545) during the sampling period. When the conversion process starts, the SAR control logic and charge redistribution DAC are used to add and subtract fixed amounts of charge from the sampling capacitor to bring the comparator into a balanced condition. When the comparator is balanced, the conversion is complete and the ADC output code is generated.

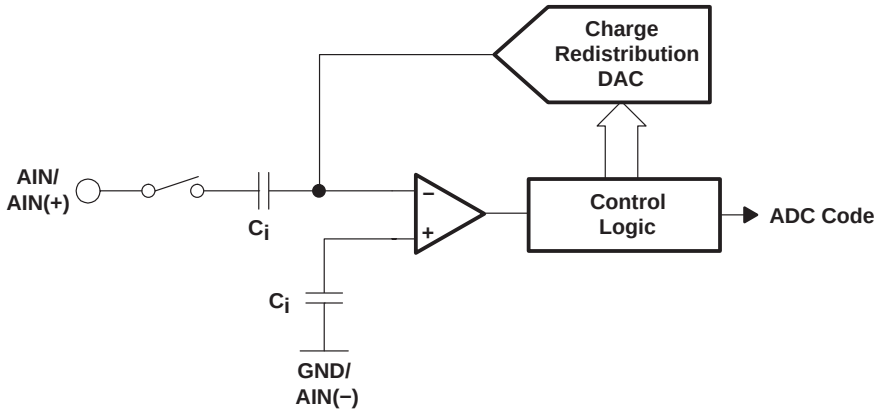


Figure 22. Simplified SAR Circuit

PRINCIPLES OF OPERATION

pseudo-differential inputs

The TLC4545 operate in pseudo-differential mode. The inverted input is available on pin 5. The inverted input can tolerate a maximum input ripple of ±0.2 V. It is normally used for zero-scale offset cancellation or ground noise rejection.

serial interface

Output data format is binary (unipolar straight binary).

binary

- Zero Scale Code = 0000h, V_{AIN} = GND
- Full Scale Code = FFFFh, $V_{AIN} = V_{REF} – 1LSB$

reference voltage

An external reference must be applied via pin 2, V_{REF}. The voltage level applied to this pin establishes the upper limit of the analog inputs to produce a full-scale reading. The value of V_{REF}, and the analog input should not exceed the positive supply or be less than GND, consistent with the specified absolute maximum ratings. The digital output is at full scale when the input signal is equal to or higher than V_{REF} and at zero when the input signal is equal to or less than GND.

auto-power down and power up

Auto-power down is built into the devices in order to reduce power consumption. The wake-up time is fast enough to provide power down between each conversion cycle. The power down state is initiated at the end of conversion and wakes up on a falling $\overline{CS}$ edge (or rising FS edge, whichever occurs first, for TLC4541 only).

APPLICATION INFORMATION

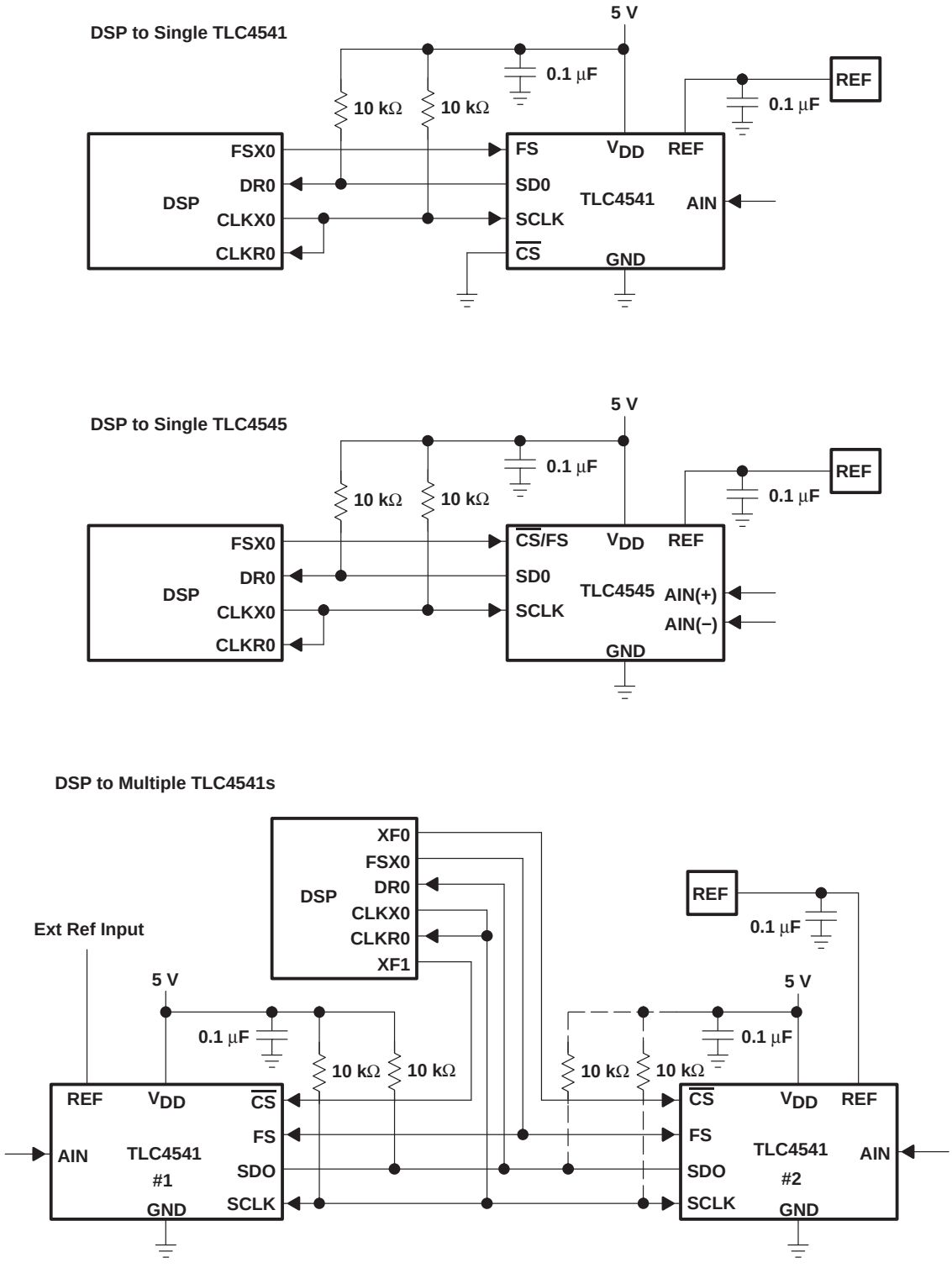


Figure 23. Typical ADC Interface to a TMS320 DSP

PACKAGING INFORMATION

Orderable part number	Status (1)	Material type (2)	Package | Pins	Package qty | Carrier	RoHS (3)	Lead finish/ Ball material (4)	MSL rating/ Peak reflow (5)	Op temp (°C)	Part marking (6)
TLC4541ID	Active	Production	SOIC (D) | 8	75 | TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	4541I
TLC4541ID.B	Active	Production	SOIC (D) | 8	75 | TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	4541I
TLC4541IDG4	Active	Production	SOIC (D) | 8	75 | TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	4541I
TLC4541IDG4.B	Active	Production	SOIC (D) | 8	75 | TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	4541I
TLC4541IDGK	Active	Production	VSSOP (DGK) | 8	80 | TUBE	Yes	NIPDAUAG	Level-1-260C-UNLIM	-40 to 85	ALM
TLC4541IDGK.B	Active	Production	VSSOP (DGK) | 8	80 | TUBE	Yes	NIPDAUAG	Level-1-260C-UNLIM	-40 to 85	ALM
TLC4541IDGKR	Active	Production	VSSOP (DGK) | 8	2500 | LARGE T&R	Yes	Call TI | Nipdauag	Level-1-260C-UNLIM	-40 to 85	ALM
TLC4541IDGKR.B	Active	Production	VSSOP (DGK) | 8	2500 | LARGE T&R	Yes	Call TI	Level-1-260C-UNLIM	-40 to 85	ALM
TLC4545ID	Active	Production	SOIC (D) | 8	75 | TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	4545I
TLC4545ID.B	Active	Production	SOIC (D) | 8	75 | TUBE	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	4545I
TLC4545IDGK	Active	Production	VSSOP (DGK) | 8	80 | TUBE	Yes	Call TI | Nipdauag	Level-1-260C-UNLIM	-40 to 85	AME
TLC4545IDGK.B	Active	Production	VSSOP (DGK) | 8	80 | TUBE	Yes	Call TI	Level-1-260C-UNLIM	-40 to 85	AME
TLC4545IDR	Active	Production	SOIC (D) | 8	2500 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	4545I
TLC4545IDR.B	Active	Production	SOIC (D) | 8	2500 | LARGE T&R	Yes	NIPDAU	Level-1-260C-UNLIM	-40 to 85	4545I

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

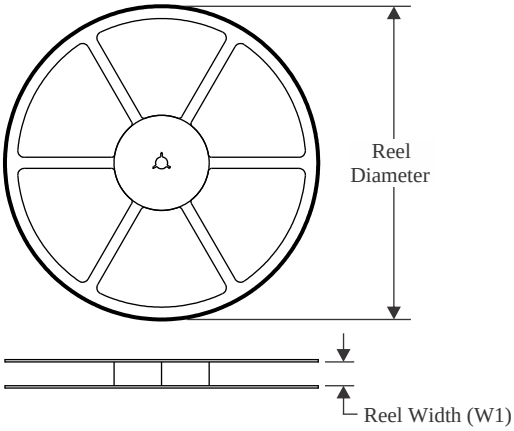
Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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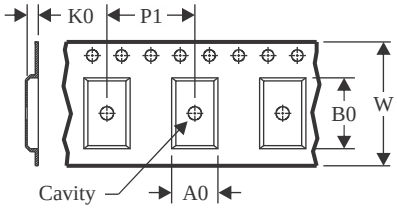
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TAPE AND REEL INFORMATION

REEL DIMENSIONS

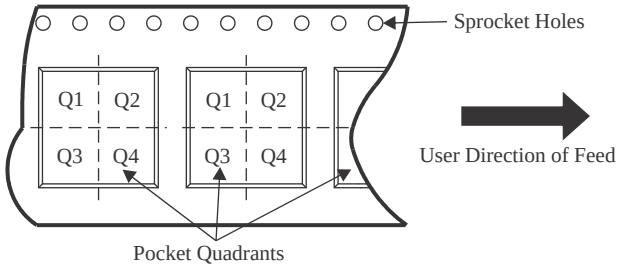


TAPE DIMENSIONS



A0	Dimension designed to accommodate the component width
B0	Dimension designed to accommodate the component length
K0	Dimension designed to accommodate the component thickness
W	Overall width of the carrier tape
P1	Pitch between successive cavity centers

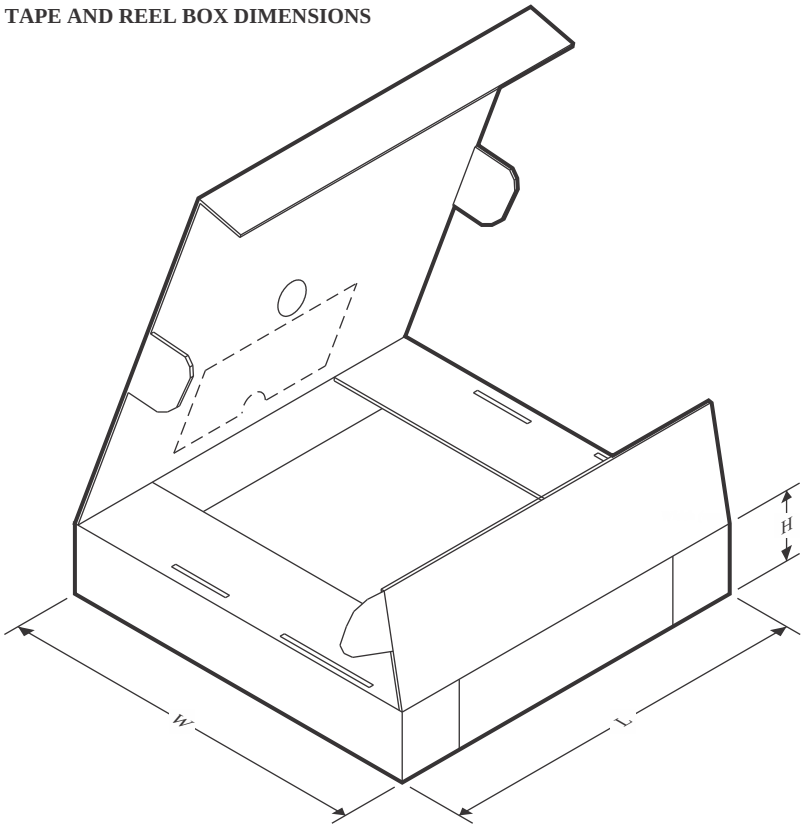
QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE



*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
TLC4545IDR	SOIC	D	8	2500	330.0	12.4	6.4	5.2	2.1	8.0	12.0	Q1

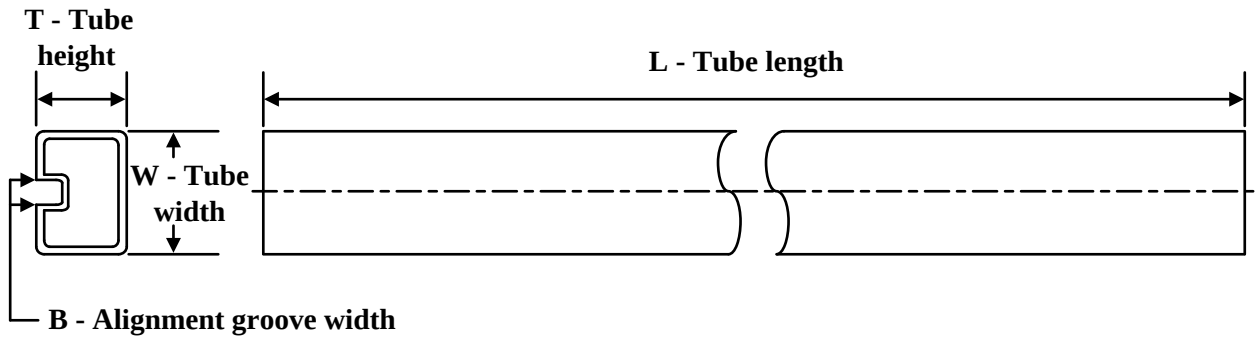
TAPE AND REEL BOX DIMENSIONS



*All dimensions are nominal

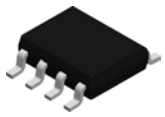
Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
TLC4545IDR	SOIC	D	8	2500	350.0	350.0	43.0

TUBE



*All dimensions are nominal

Device	Package Name	Package Type	Pins	SPQ	L (mm)	W (mm)	T (µm)	B (mm)
TLC4541ID	D	SOIC	8	75	505.46	6.76	3810	4
TLC4541ID.B	D	SOIC	8	75	505.46	6.76	3810	4
TLC4541IDG4	D	SOIC	8	75	505.46	6.76	3810	4
TLC4541IDG4.B	D	SOIC	8	75	505.46	6.76	3810	4
TLC4541IDGK	DGK	VSSOP	8	80	331.47	6.55	3000	2.88
TLC4541IDGK.B	DGK	VSSOP	8	80	331.47	6.55	3000	2.88
TLC4545ID	D	SOIC	8	75	505.46	6.76	3810	4
TLC4545ID.B	D	SOIC	8	75	505.46	6.76	3810	4
TLC4545IDGK	DGK	VSSOP	8	80	331.47	6.55	3000	2.88
TLC4545IDGK.B	DGK	VSSOP	8	80	331.47	6.55	3000	2.88

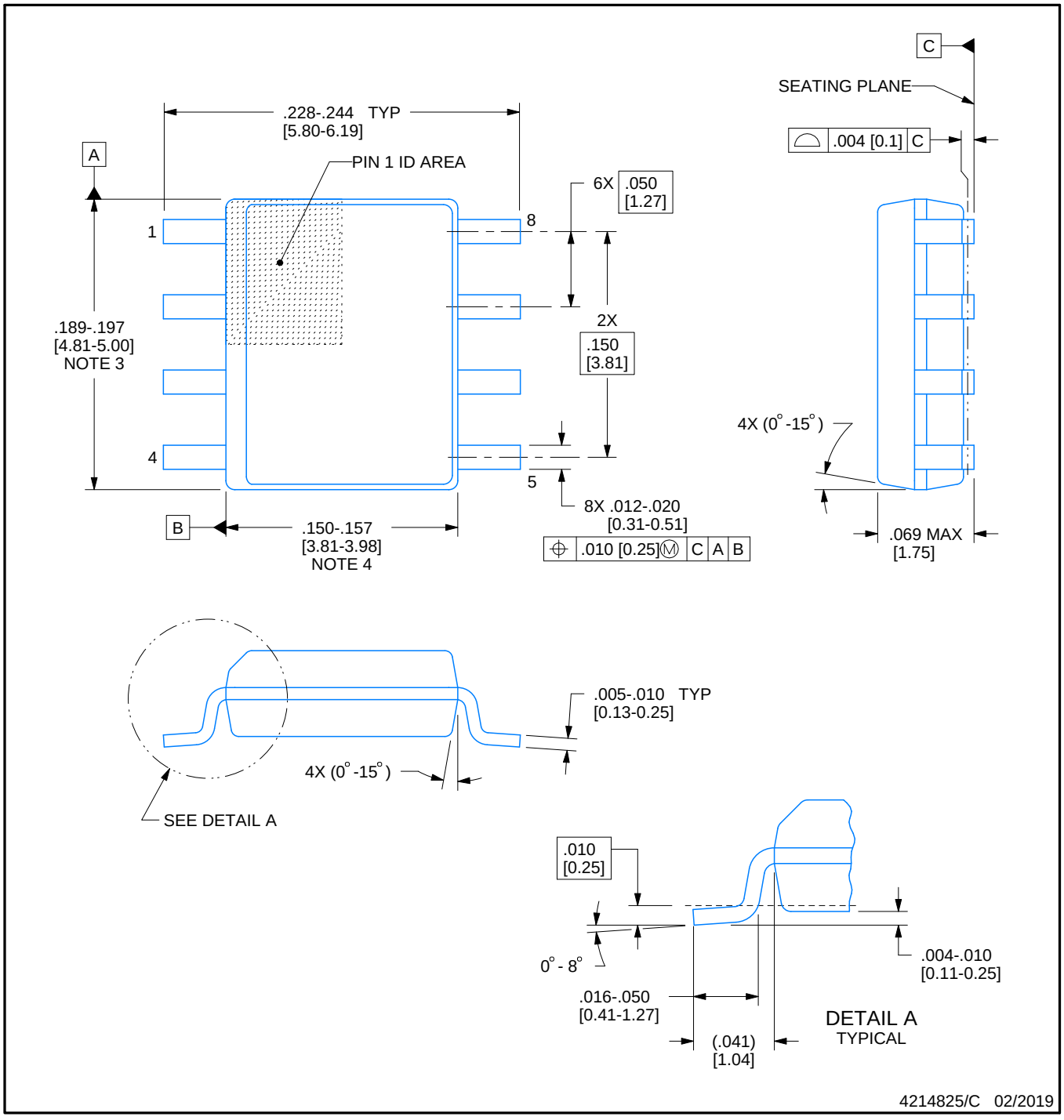


PACKAGE OUTLINE

D0008A

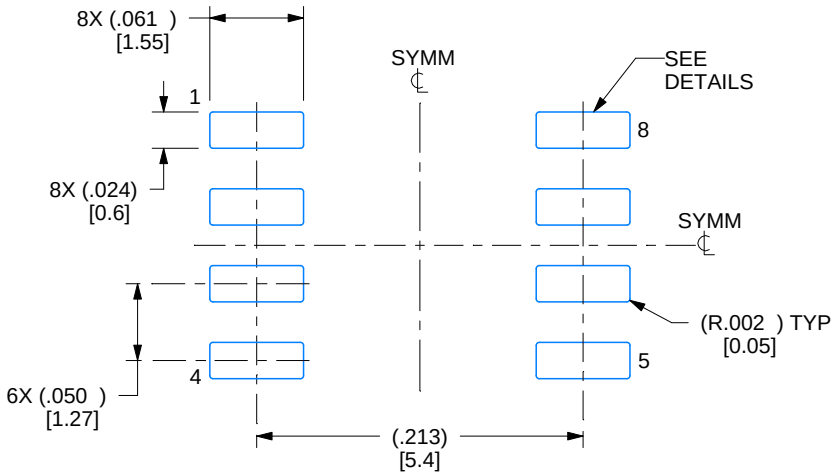
SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT

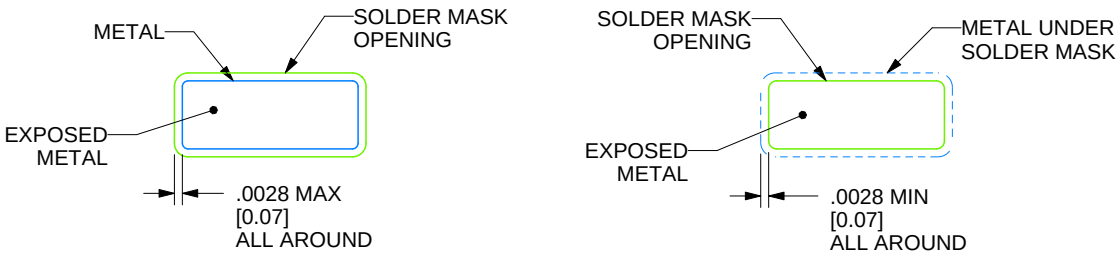


NOTES:

1. Linear dimensions are in inches [millimeters]. Dimensions in parenthesis are for reference only. Controlling dimensions are in inches. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed .006 [0.15] per side.
4. This dimension does not include interlead flash.
5. Reference JEDEC registration MS-012, variation AA.



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE:8X



SOLDER MASK DETAILS

4214825/C 02/2019

NOTES: (continued)

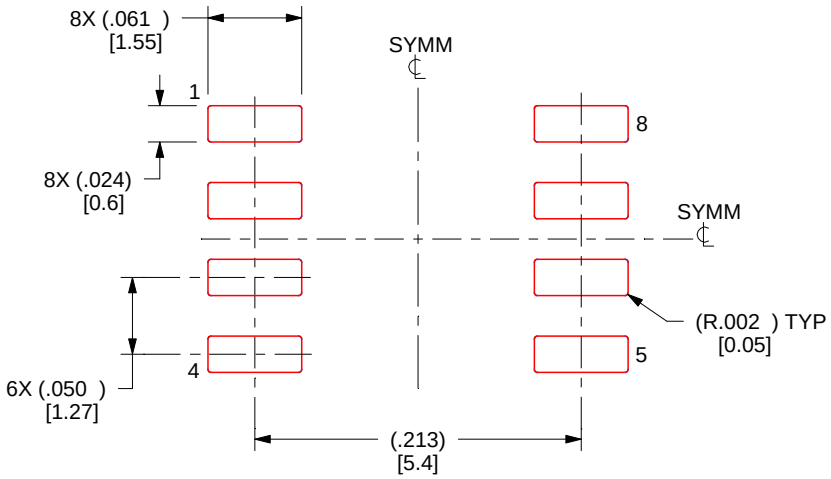
6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

EXAMPLE STENCIL DESIGN

D0008A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



SOLDER PASTE EXAMPLE
BASED ON .005 INCH [0.125 MM] THICK STENCIL
SCALE:8X

4214825/C 02/2019

NOTES: (continued)

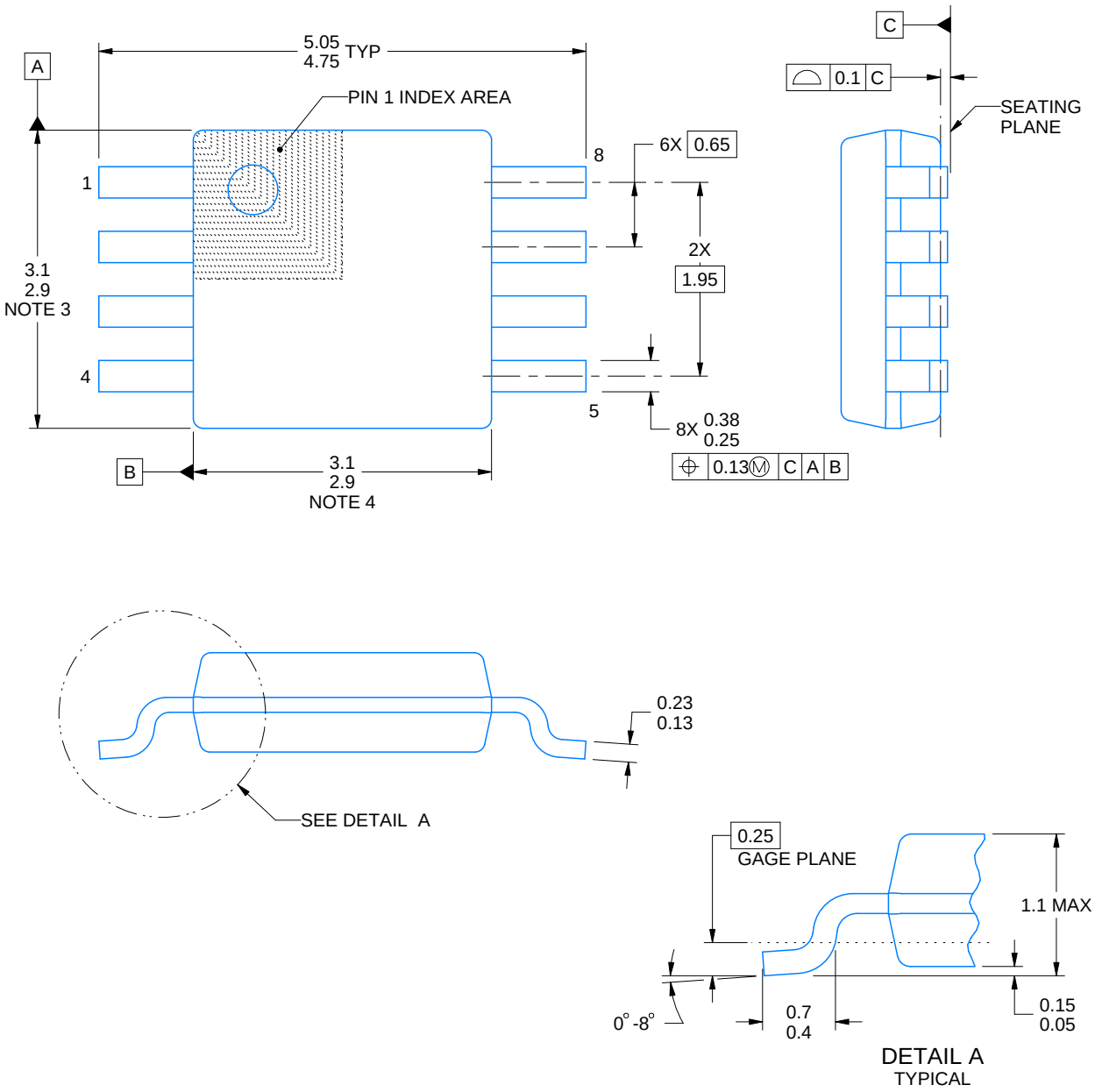
8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

PACKAGE OUTLINE

DGK0008A

VSSOP - 1.1 mm max height

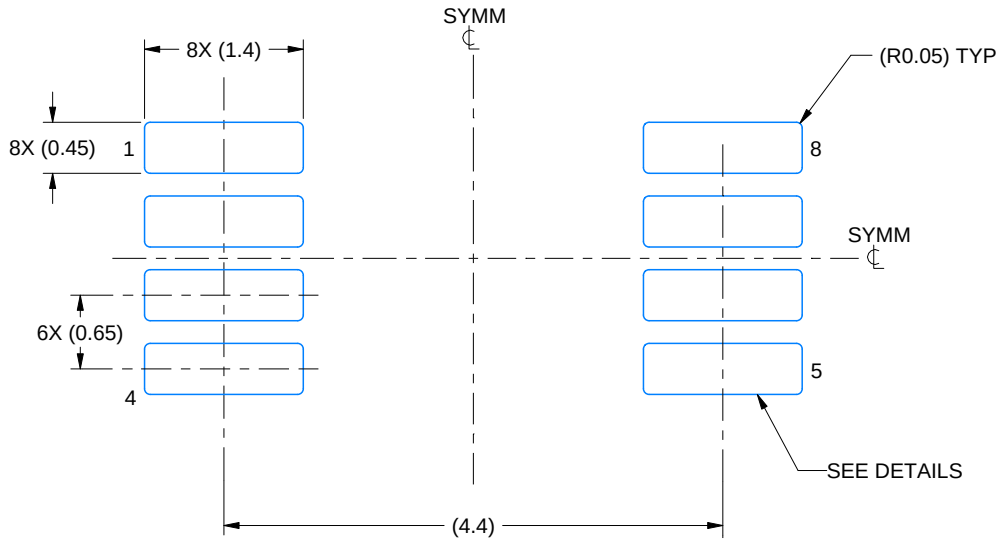
SMALL OUTLINE PACKAGE



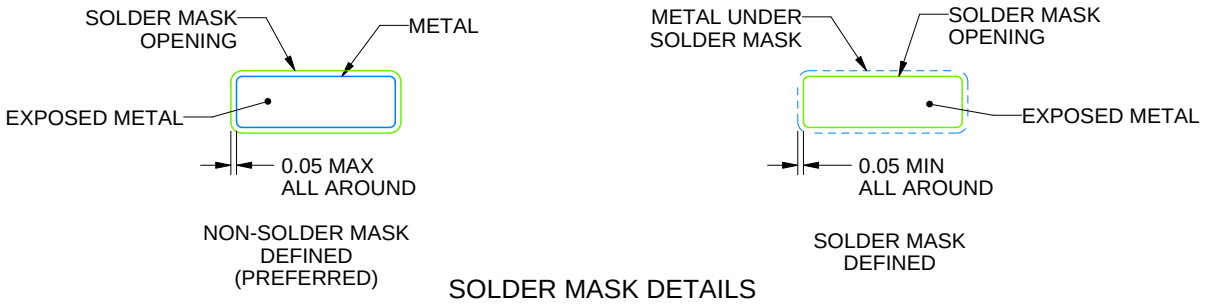
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NOTES:

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2. This drawing is subject to change without notice.
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4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.25 mm per side.
5. Reference JEDEC registration MO-187.



LAND PATTERN EXAMPLE
EXPOSED METAL SHOWN
SCALE: 15X

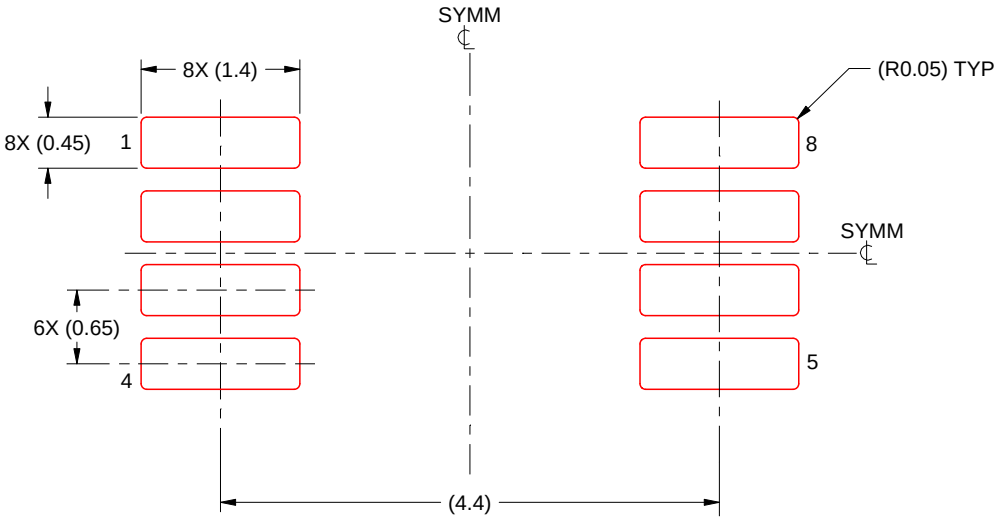


SOLDER MASK DETAILS

4214862/A 04/2023

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.
8. Vias are optional depending on application, refer to device data sheet. If any vias are implemented, refer to their locations shown on this view. It is recommended that vias under paste be filled, plugged or tented.
9. Size of metal pad may vary due to creepage requirement.



SOLDER PASTE EXAMPLE
SCALE: 15X

4214862/A 04/2023

NOTES: (continued)

11. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
12. Board assembly site may have different recommendations for stencil design.

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